

Alireza Laalmanesh

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Academic Background

University of Waterloo

Master of Mathematics (MMath) in Computational Mathematics (Co-op)

Waterloo, Canada

Sep. 2025 – Present

- Grateful for the patient mentorship of **Dr. David Del Rey Fernández** and **Dr. Leia Minaker**.

Amirkabir University of Technology (Tehran Polytechnic)

B.Sc. in Industrial Engineering

Tehran, Iran

Sep. 2020 – Jun. 2025

- Academic Standing: **A Average** (Focus on Systems Analysis and Mathematical Modeling).
- Thesis: "Investigating Operating Room Scheduling via Meta-heuristic Optimization".

Knowledge Journey & Tools

Languages: Python, C/C++ (Tools for translating Math into Logic)

AI Explorations: PyTorch, TensorFlow, CNNs, DeepInsight Methodology, Discrete Event Simulation (DES)

Systems & Automation: Docker, Linux (low-level utilities), ADB protocols, Selenium, Asyncio scripts

Interests: Numerical Optimization, Stochastic Modeling, Custom Loss Function Experiments

Professional Experience

University of Waterloo

Graduate Teaching Assistant

Waterloo, Canada

Sep. 2025 – Present

- Current Term: Facilitating tutorials for **MATH 128** (Calculus 2 for Sciences) and **MATH 106** (Applied Linear Algebra 1).
- Previous Term: Assisted with **MATH 135** (Algebra for Honours Math) and **MATH 115** (Linear Algebra for Engineering).
- Supporting students in mastering fundamental logic and mathematical concepts through regular feedback.

Amirkabir University of Technology

Undergraduate Teaching Assistant

Tehran, Iran

Sep. 2023 – Jun. 2025

- Assisted in "Optimization" and "Systems Analysis" courses under Dr. Aminnayeri and Dr. Mosadegh.
- Drafted modeling assignments to help peers bridge theory with practical engineering applications.

TAV Institute (Tavenergyco)

Factory Layout Design Intern

Tehran, Iran

Jul. 2023 – Dec. 2023

- Applied theoretical optimization to real-world layout challenges, focusing on process flow efficiency.
- Learned to identify production bottlenecks through capacity analysis and spatial modeling.

Recent Experiments

Traffic Severity Study | *PyTorch, Custom Loss Design*

- Studying imbalanced data challenges in traffic patterns using custom **Soft F-Beta Loss** functions.
- Experimenting with minority class representation to improve model sensitivity to fatal accident data.

CrashVision DL Exploration | *Computer Vision, DeepInsight*

- Practicing feature mapping techniques by converting heterogeneous data into images via **DeepInsight**.
- Exploring CV-based approaches for severity prediction as part of a broader learning journey in AI.

LinguoMath TTS Experiment | *Python, Asyncio, Regex*

- Investigating ways to parse complex L^AT_EX formulas from scientific PDFs into natural language.
- Built an experimental pipeline using **Asyncio** to learn about real-time bilingual audio generation.

Android Hardware Bridge | *Low-level Protocols, ADB*

- A personal project exploring **USB/IP protocols** to repurpose rooted Android devices as headless servers.
- Learned about Linux hardware interfaces and TCP/IP networking by bypassing standard sandboxing.

CodeBase Aggregator | *Automation, File I/O*

- Developed a utility to aggregate repositories into datasets for better LLM context injection studies.
- Practiced Python automation and efficient file I/O handling for dataset curation.

Foundations & Recognition

Ranked in the **Top 1%** of candidates in the National University Entrance Exam (+150,000 participants).

Recipient of a merit-based scholarship for academic progress at Amirkabir University.

Self-employed Tutor: Dedicated over 5 years to fostering deep conceptual understanding in Math and Physics for high school students.